Fiscal Year:	FY 2026	**Task Last Updated:**	FY 05/01/2026
PI Name:	Jain, Abhishek Ph.D.		
Project Title:	Long-term Patient iPSC Vessel Chip Model to Assess Stressors of Atherosclerosis and mRNA Therapeutics		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:	NIH, BARDA, FDA	**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	(1) Cell Culture (2) Translational (Countermeasure) Potential		
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Zip Code:	77843	**Congressional District:**	17
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2021 Space Biology NNH21ZDA015N. Extended Longevity of 3D Tissues and Microphysiological Systems for Modeling of Acute and Chronic Exposures to Stressors
Start Date:	03/20/2023	**End Date:**	02/20/2027
No. of Post Docs:	2	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	1
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	3	**Monitoring Center:**	NASA HQ
Contact Monitor:	Koniges, Ursula	**Contact Phone:**	202-256-8786
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Flight Program:			
Flight Assignment:	Per the Research Announcement for this solicitation: this effort is a collaboration between NASA; the National Institutes of Health (NIH); the Biomedical Advanced Research and Development Authority (BARDA); and the Food and Drug Administration (FDA).		
Key Personnel Changes/Previous PI:	Dr. Vladislav Yakovlev (coI) is no longer associated with the project. Dr. Jain is responsible to complete the tasks associated with him. Per the FY26 report, Nhat Tu Le is no longer included in the list of Co-Investigators (Ed., 5/28/26).		
COI Name (Institution):	Wang, Guangyu Ph.D. (Houston Methodist Hospital) Cooke, John P M.D., Ph.D. (The Houston Methodist Research Institute)		
Grant/Contract No.:	80ARC023CA002		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Animals are not very precise modeling systems of human physiology and cannot ensure dissectible signaling analysis. Recent advances in organ-on-chip technology or microphysiological systems (MPS) have permitted co-culture of human cells in physiologically-relevant microfluidic environments and also allowed continuous perfusion, providing an alternative in vitro approach to mimic the human disease. These microsystems, however, still have not reached their full potential and made in-depth entry into informing clinical trials due to the absence of efficient and scalable primary human patient cells, limited real-time biosensing potential, and critical validation over long durations. Amongst all human disease conditions, vascular diseases can be often fatal in which symptoms such as aggressive atherosclerosis exist. Amongst other risk factors, age is one of the most critical. Radiation exposure during space travel is another. Therefore, investigating vascular signaling mechanisms and finding therapeutics for aging and radiation-related vascular diseases is of high prospective impact. But there is a fundamental gap in the understanding of: 1) how organ-level signaling occurs in blood vessels amongst atherosclerotic patients, and 2) how cells inside the vascular organ contribute to the disease and respond to drugs over time. To meet these complex challenges, the team's goal is to take Progeria patients and proton radiation as case-studies and demonstrate that 3D cylindrical vessel-chips containing co-culture of vascular endothelial and smooth muscle cells from human induced pluripotent stem cells (hiPSCs) will provide real-time readouts of critical vascular and hematological biomarkers, that inform clinical outcomes. The chip will consist of physiologically-relevant extracellular matrix that will be continually monitored and it will also be ready for microgravity experiments that can be done in space. The objective is to establish long-term culture and feasibility of a Vessel-Chip system that helps dissect pathophysiology of atherosclerosis and toxin/drug-tissue interactions with much higher specificity and sensitivity than gold-standard models. We will add more advanced biosensing to our existing Vessel-Chip that mimics vessel geometry, hemodynamics, and extracellular matrix, to continuously monitor physiological parameters and sustain the culture of human iPSC-derived endothelial cells (hiPSC-EC) and vascular smooth muscle cells (hiPSC-SMC) for 180 days from healthy patients (Phase I). We will assess HGPS patient iPSC vascular cells and protons as biological and space-relevant stressors respectively over 180 days, and finally, novel Good Manufacturing Practice (GMP)-grade mRNA therapeutics loaded in a unique lipid nanoparticle-based delivery vehicle will be evaluated through a panoply of organ-to-gene level assays, including clinically-relevant biomarkers (Phase II). The research proposed is innovative and significant, because it contributes in shifting the status quo of vascular pathophysiology in the organ-on-a-chip platform by containing primary hiPSC-vascular cells, blood flow, and associated transcriptomics and proteomics over long durations of co-culture and continuous monitoring. The outcomes will lead to further applications of the organ-on-chip technology in translational research in atherosclerosis and other vascular and hematological diseases.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Potential Impact: -- A long-duration microphysiological model of atherosclerosis. -- New knowledge of aging and radiation-induced vascular disease. -- A testbed for mRNA therapeutic preclinical trials. -- Enabling future experimentation at the International Space Station (ISS) under microgravity.
Task Progress:	We have progress in 3 technical areas, as follows: 1. We engineered and maintained a human vascular avatar for over 180 days by using an organ-on-chip (Vessel-Chip) model which exposes human vascular cells to an optimized culture media containing factors that significantly enhance their long-term health. This long-term human vascular avatar. which incorporates assessment of clinically-relevant markers of human vascular cells in a compositionally and geometrically relevant matrix for several months, is now expected to be an extraordinarily useful tool for studying vascular disease pathobiology and will facilitate the drug discovery by allowing longitudinal assessment of vascular function relevant to chronic disease or space travel. 2. A crucial factor limiting the longevity and ease of adoption of microphysiological systems (MPS) is the constrained non-specific design and performance of existing perfusion systems. Inspired by the physics of the accordion music instrument, we have engineered Hemadyne, a standalone mechanical pump with a compact footprint that operates without any additional instruments. It is paired with a custom-built control algorithm to reproduce clinical waveforms with a 400-millisecond temporal resolution and replicates Doppler ultrasound waveforms of patient hemodynamics with nearly absolute spatiotemporal fidelity. Surpassing conventional technologies, Hemadyne provides sustainable and stable flow, including transient forward and backward flows, comprising multiple amplitudes and phases within a single period, with high signal-to-noise ratio and extremely fast response times. Consequently, Hemadyne sustained the long-term culture of primary human endothelial cells in a vessel-chip for up to 60 days. It also dissected the role of hemodynamic diastolic rest phase as a determinant of endothelial homeostasis, while revealing a differential response to arterial and venous endothelial cells. Finally, it described the age-associated pathological effect of transient diastolic retrograde flow (flow reversal) on arterial endothelial cells, that animal models cannot reproduce. Therefore, Hemadyne is an enabling technology that improves operational longevity and physiological-relevance of MPS. 3. Cell-culture incubators are indispensable for in vitro research and new approach methods (NAMs), yet they remain vulnerable to undetected environmental fluctuations that degrade experimental reproducibility and compromise biological sample integrity over long-term. Existing monitoring solutions partially mitigate this risk but are frequently encumbered by wired infrastructure, external signal-transmitter assemblies, or a large form factor that hinders integration with diverse workflows. This study presents IncuSense, a miniaturized, battery-powered, wireless environmental monitoring platform capable of sensing temperature, humidity, and CO_2 in real-time. The system integrates a photoacoustic non-dispersive infrared (NDIR) sensor with a microcontroller, enabling periodic data acquisition and transmission to local or cloud-hosted dashboards for real-time visualization and automated alerts. Comprehensive thermal characterization identified a critical limitation of continuous microcontroller unit (MCU) operation: Wi-Fi standby power consumption induced progressive self-heating, yielding an internal temperature rise of approximately 18°C and systematic bias in temperature and humidity readouts. To address this challenge, an intermittent power-cycling strategy was implemented to gate power to all ESP32 subsystems except the real-time clock, reducing the average current draw by more than 90-fold to 0.92 mA, suppressing thermal drift, and extending operational lifetime. Benchmarking against National Institute of Standards and Technology (NIST)-traceable sensors demonstrated high accuracy and precision in environmental measurements, and incubator failure recording. 100-day deployment in a CO_2 incubator further confirmed long-term stability, with minimal drift and demonstrated its ability to detect transient fluctuations caused by routine incubator access. Collectively, these results establish IncuSense as a compact, cost-effective, and adaptable platform for continuous incubator monitoring, facilitating environmental

	standardization in long-term culture workflows, and may support the standardization of NAMs.
Bibliography Type:	Description: (Last Updated: 06/02/2026)
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